

Program : Diploma in Computer Science and Engineering	
Course Code: 5408	Course Title: Artificial Intelligence Lab
Semester: 5	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

Students will be able to solve basic and application level problems in Artificial Intelligence.

Course Prerequisites:

Topic	Course Name
Basic Knowledge in Computer Science	Introduction to IT Systems Lab
Basic Computer programming knowledge	Problem Solving and Programming

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Develop solutions to basic problems in AI using Prolog.	9	Applying
CO2	Implement solutions to different AI algorithms using Python	12	Applying
CO3	Implement different AI problems using Python	12	Applying
CO4	Open Ended Experiment	9	Applying
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3	3	3	3	3	3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of the Experiment	Duration (Hours)	Cognitive Level
CO1	Design solutions to basic problems in AI using Prolog		
M1.01	Practice installation of Prolog	3	Applying
M1.02	Write simple facts for the statement using Prolog	3	Applying
M1.03	Implement simple arithmetic in Prolog	3	Applying
CO2	Implement solutions to different AI algorithms using Python		
M2.01	Solve any mathematical problem using DFS	3	Applying
M2.02	Solve any mathematical problem using BFS	3	Applying
M2.03	Demonstrate A* algorithm	3	Applying
M2.04	Demonstrate Hill Climbing algorithm	3	Applying
	Lab Exam -1	1.5	
CO3	Implement different AI problems using python		
M3.01	Implementation of Tower of Hanoi	3	Applying
M3.02	Solve four-queen's problem.	3	Applying
M3.03	Implement Travelling Salesman Problem	3	Applying
M3.04	Implement a water-jug problem	3	Applying
CO4	Open Ended Experiments**		
	Lab Exam – II	1.5	

**** - Suggested Open Ended Experiments**

(Not for End Semester Examination but compulsory to be included in Continuous Internal

Evaluation. Students can do open-ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

1. Solve any of the following AI problems
 - a. Eight puzzle problem
 - b. Eight Queens Problem
 - c. Chess Problem
 - d. Cryptarithmic Puzzles
2. Implement any one of the following Expert system like MYCIN , E MYCIN etc
 - a. Information management
 - b. Hospitals and medical facilities
 - c. Help desks management
 - d. Employee performance evaluation
 - e. Stock market trading

Text / Reference

T/R	Book Title/Author
T1	S. Russell and P. Norvig, "Artificial Intelligence: A Modern Approach, Prentice Hall, Third Edition, 2009..
T1	Artificial Intelligence: A Modern Approach, 4th Edition, Stuart Russell, peter Norvig University of California at Berkeley, Pearson education, 2020.
T2	I. Bratko, —Prolog: Programming for Artificial Intelligence, Fourth Edition, Addison-Wesley Educational Publishers Inc., 2011
R1	M. Tim Jones, —Artificial Intelligence: A Systems Approach (Computer Science),
R3	Jones and Bartlett Publishers, Inc.; First Edition, 2008 3. Nils J. Nilsson, —The Quest for Artificial Intelligence, Cambridge University ,2009

Online Resources

Sl.No	Website Link
1	An Introduction to Artificial Intelligence - Course (nptel.ac.in)
2	Fundamentals Of Artificial Intelligence - Course (nptel.ac.in)